HALLUCINATION AUDIT OF AI CHATBOTS

Mini Project Report

Project Domain: Artificial Intelligence and Data Science

Prepared by: _____

Department / Class: _____

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A beginner-friendly project to record, review, and analyse factual errors in AI chatbot responses.

1. Abstract

AI chatbots can sometimes produce incorrect, misleading, or unsupported information in a confident style. This is commonly called an AI hallucination. This project presents a simple dashboard for auditing chatbot answers. The user records a question, chatbot response, reference answer, source, and reviewer label. The application summarises the results, calculates a hallucination rate for verifiable records, and displays charts. The tool supports structured human review; it does not independently guarantee factual truth.

2. Introduction

Chatbots are used for learning, research, customer support, and content creation. Because their responses may contain factual errors, users need a repeatable way to inspect them. A hallucination audit evaluates answers against reliable evidence and records the reasoning behind each decision.

3. Problem Statement

There is a need for a simple, beginner-friendly method to organise chatbot answers and measure how often evaluated answers contain incorrect or unsupported claims. Manually checking scattered answers can make comparisons difficult. This project provides a central table and summary dashboard.

4. Objectives

• Collect chatbot answers for a set of questions.
• Compare each answer with a verified reference or expected facts.
• Label answers as Correct, Partially Correct, Incorrect / Hallucinated, or Unverifiable.
• Calculate and display the hallucination rate for verifiable records.
• Export the audit dataset for further analysis and reporting.

5. Scope

The project is suitable for a small academic experiment using factual questions from science, technology, history, and general knowledge. It can compare responses from different chatbots when the same questions and evaluation rules are used. Results depend on the quality of the references and the consistency of human reviewers.

6. Tools and Technologies

Python is used for application logic; Streamlit builds the web interface; Pandas manages the audit table; Matplotlib creates a chart; CSV stores the dataset; and VS Code is the recommended editor. A web browser is used to view the dashboard.

7. System Modules

Question and response entry: record the question, chatbot name, and exact response.
Reference management: record expected facts and a source for checking.
Evaluation: a reviewer chooses a label after inspecting the evidence.
Analysis: count each label and calculate rates.
Visualisation and export: display charts and download results as CSV.

8. Methodology

1. Prepare a fixed list of factual questions.
2. Ask the same questions to each chatbot under similar conditions.
3. Save the exact responses and note the chatbot/model and date if available.
4. Verify claims using trustworthy sources such as official documentation, academic sources, or reputable

reference works.
5. Apply the same label definitions to each response.
6. Review summary statistics and discuss limitations.

9. Evaluation Labels

Correct: the answer is supported by the reference evidence.
Partially Correct: some claims are right, but important details are missing or misleading.
Incorrect / Hallucinated: a material factual claim contradicts reliable evidence or is presented without support as fact.
Unverifiable: the available evidence is insufficient to decide. This label should not automatically be treated as a hallucination.

10. Formula

Hallucination Rate (%) = (Number of Incorrect / Hallucinated answers ÷ Number of verifiable answers) × 100. Verifiable answers include Correct, Partially Correct, and Incorrect / Hallucinated records. Unverifiable records are excluded from this denominator and reported separately. The report should state its label definitions and denominator clearly.

11. Sample Dataset

The included CSV contains five illustrative examples. They demonstrate how to use the application and are not results from a live chatbot experiment. Replace or supplement them with answers you actually collect and verify before submitting the final project.

12. Expected Output

The dashboard displays the number of records, correct answers, incorrect/hallucinated answers, hallucination rate, evaluation-label chart, and incorrect-answer counts by chatbot. Users can download all audit records as a CSV file.

13. Testing

Test the app by adding a complete record, leaving required fields blank, selecting each evaluation label, downloading the CSV, and checking whether the displayed counts update correctly. Verify that unverifiable answers are excluded from the hallucination-rate denominator.

14. Limitations

The dashboard does not automatically fact-check answers. Human reviewers may disagree, sources may be incomplete, and model outputs may change over time. A small question set cannot represent every task or user. For stronger research, use a larger balanced dataset, independent reviewers, documented sources, and repeated trials.

15. Future Enhancements

Possible improvements include a larger benchmark dataset, two independent reviewers, source-quality scoring, model/version tracking, automated retrieval of reference evidence, and carefully validated natural-language processing assistance. Any automated scoring should be checked against human-labelled examples.

16. Conclusion

The Hallucination Audit of AI Chatbots project offers a practical way to organise chatbot responses and evaluate them consistently. Its dashboard and charts help users identify patterns in factual errors. The project emphasises evidence-based review and transparent reporting rather than assuming that automated scores are

always correct.

17. References

• Python documentation: https://www.python.org/doc/
• Streamlit documentation: https://docs.streamlit.io/
• Pandas documentation: https://pandas.pydata.org/docs/
• Matplotlib documentation: https://matplotlib.org/stable/
• Reference sources should be selected for each factual question and recorded in the audit dataset.

Appendix: How to Run

1. Install Python 3.10 or newer and VS Code.
2. Open the project folder in VS Code.
3. In the terminal, run `pip install -r requirements.txt`.
4. Run `streamlit run app.py`.
5. Open the local address shown in the terminal. The included sample data appears in the dashboard.